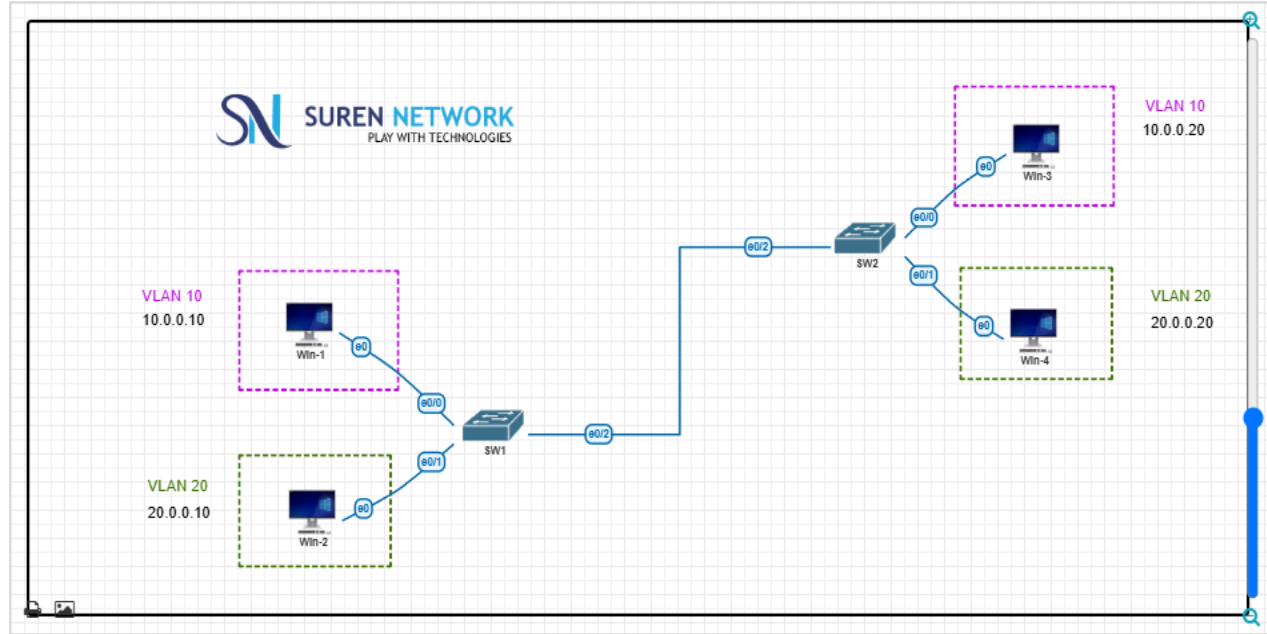


|                          |                               |                   |
|--------------------------|-------------------------------|-------------------|
| Search Labs              |                               |                   |
| <div>..</div>            |                               |                   |
| <input type="checkbox"/> | 1-LAB VLAN NEW.unl            | 22 Dec 2025 16:48 |
| <input type="checkbox"/> | 4-LAB INTERVLAN.unl           | 17 Oct 2025 07:02 |
| <input type="checkbox"/> | 5-LAB HSRPFLOATING STATIC.unl | 04 Sep 2025 09:39 |
| <input type="checkbox"/> | 3-LAB ETHERCHANNEL.unl        | 04 Sep 2025 09:39 |
| <input type="checkbox"/> | 2-LAB STP.unl                 | 04 Sep 2025 09:39 |

## CCNA CORE TOPIC

### Switching Part – I LAB

#### Lab Preview

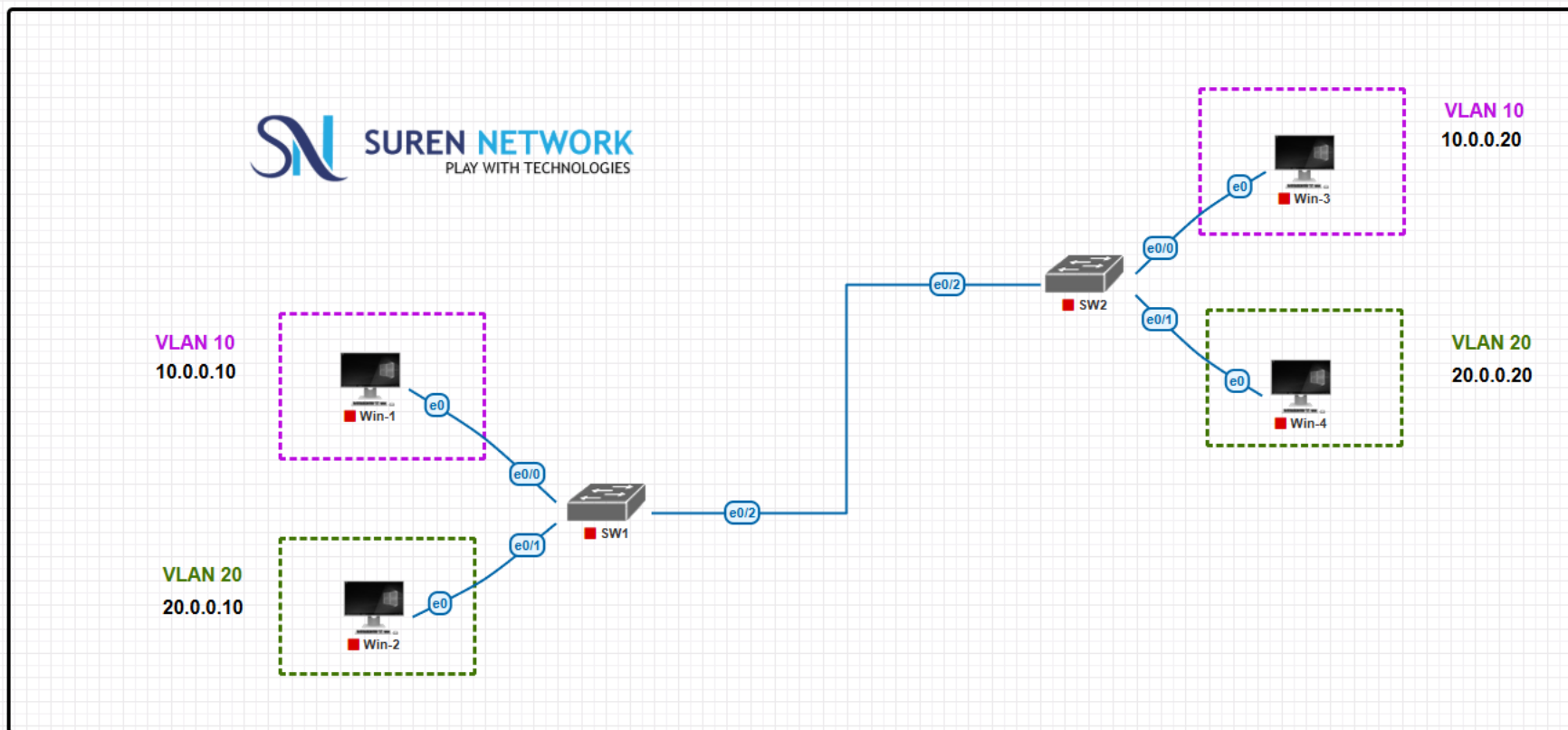


Open Edit Delete

- Lab Name : 1-LAB VLAN NEW
- Author : 1
- Version : 1
- Lab ID : 84bc8572-9fe3-455a-82e3-041af75ab638
- Description :

Class date: 15.09.2026

By Meganadhan



Win-1 ✕ Win-2 ✕ Win-3 ✕ Win-4 ✕ SW1 ✕

```
SW1#show vlan brief
```

| VLAN | Name               | Status    | Ports                      |
|------|--------------------|-----------|----------------------------|
| 1    | default            | active    | Et0/0, Et0/1, Et0/2, Et0/3 |
| 1002 | fddi-default       | act/unsup |                            |
| 1003 | token-ring-default | act/unsup |                            |
| 1004 | fddinet-default    | act/unsup |                            |
| 1005 | trnet-default      | act/unsup |                            |

```
SW1#config t
```

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

```
SW1(config)#vlan 10
```

```
SW1(config-vlan)#name NCCR
```

```
SW1(config-vlan)#ex
```

```
SW1(config)#vlan 20
```

```
SW1(config-vlan)#name NIOT
```

```
SW1(config-vlan)#ex
```

```
SW1(config)#do sh vlan br
```

| VLAN | Name               | Status    | Ports                      |
|------|--------------------|-----------|----------------------------|
| 1    | default            | active    | Et0/0, Et0/1, Et0/2, Et0/3 |
| 10   | NCCR               | active    |                            |
| 20   | NIOT               | active    |                            |
| 1002 | fddi-default       | act/unsup |                            |
| 1003 | token-ring-default | act/unsup |                            |
| 1004 | fddinet-default    | act/unsup |                            |
| 1005 | trnet default      | act/unsup |                            |

```
SW1(config)#int e 0/0
```

```
SW1(config-if)#switchport access vlan 10
```

```
SW1(config-if)#exit
```

```
SW1(config)#int e 0/1
```

```
SW1(config-if)#switchport access vlan 20
```

```
SW1(config-if)#exit
```

```
SW1(config)#do sh vlan br
```

| VLAN | Name               | Status    | Ports        |
|------|--------------------|-----------|--------------|
| 1    | default            | active    | Et0/2, Et0/3 |
| 10   | NCCR               | active    | Et0/0        |
| 20   | NIOT               | active    | Et0/1        |
| 1002 | fddi-default       | act/unsup |              |
| 1003 | token-ring-default | act/unsup |              |
| 1004 | fddinet-default    | act/unsup |              |
| 1005 | trnet-default      | act/unsup |              |

```
SW1(config)#
```

## Switch 1 Configuration

```
SW1(config)#do sh vlan br
```

```
SW1(config)#int e 0/0
SW1(config-if)#switchport access vlan 10
SW1(config-if)#exit
SW1(config)#int e 0/1
SW1(config-if)#switchport access vlan 20
SW1(config-if)#exit
SW1(config)#do sh vlan br
```

```
SW1(config)#
*Sep 16 04:42:31.861: %PNP-6-PNP DISCOVERY STOPPED: PnP Discovery stopped (Config Wizard)
```

```
SW1(config)#int e 0/2
SW1(config-if)#switchport trunk encapsulation dot1q
SW1(config-if)#switchport mode trunk
SW1(config-if)#ex
SW1(config)#do sh vlan br
```

```
SW1(config)#
```

```
SW2(config)#do sh vlan br
```

```
SW2(config)#vlan 10
SW2(config-vlan)#name NCCR
SW2(config-vlan)#ex
SW2(config)#do sh vlan br
```

```
SW2(config)#int e 0/0
SW2(config-if)#switchport access vlan 10
SW2(config-if)#ex
SW2(config)#int e 0/1
SW2(config-if)#switchport access vlan 20
SW2(config-if)#ex
SW2(config)#int e 0/2
SW2(config-if)#switchport trunk encapsulation dot1q
SW2(config-if)#switchport mode trunk
SW2(config-if)#ex
SW2(config)#do sh vlan br
```

31°C Mostly sunny Search [Taskbar icons: File Explorer, Edge, Chrome, etc.] 10:21 AM 16-09-2025

```
SW2(config)#do sh vlan br
```

```
SW2(config)#int e 0/0
SW2(config-if)#switchport access vlan 10
SW2(config-if)#ex
SW2(config)#int e 0/1
SW2(config-if)#switchport access vlan 20
SW2(config-if)#ex
SW2(config)#int e 0/2
SW2(config-if)#switchport trunk encapsulation dot1q
SW2(config-if)#switchport mode trunk
SW2(config-if)#ex
SW2(config)#do sh vlan br
```

```
SW2(config)#do sh int trunk
```

```
Port      Vlans in spanning tree forwarding state and not pruned
Et0/2     1,10,20
```

```
SW2(config)#
```

```
SW1(config-if)#exit
SW1(config)#do sh vlan br
```

```

VLAN Name                Status    Ports
-----
1    default                active    Et0/3
10   NCCR                   active    Et0/0
20   NIOT                   active    Et0/1
1002 fddi-default            act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default       act/unsup
1005 trnet-default         act/unsup
SW1(config)#exit
SW1#sh
*Sep 16 04:54:01.337: %SYS-5-CONFIG_I: Configured from console by console
SW1#sh int trunk

```

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Search

Taskbar icons: File Explorer, Edge, OneDrive, Google Chrome, Search, Teams, G, OneNote, LinkedIn, Instagram, Word, Excel, PowerPoint.

System tray: 10:25 AM, 16-09-2025

PNETLab | TopologyVLAN.cdrTerminal

labs.surennetwork.com/store/public/admin/labs/terminal?terminals=eyJ0YWVJZjlpb2V2luTEILCj1cmwiOiIvaHRtbDUvYy9jbGllbnQvTXpFek16UXINREk0QUdNQWJYbHJpV3c9P3Rva2VuPTBGOTewQTIDOTQ0RDQ0OTYz...

Incognito (2)Finish update

Win-1Win-2Win-3Win-4SW1SW2

C:\WINDOWS\system32\cmd.exe

Ethernet adapter Local Area Connection:  
Connection-specific DNS Suffix . :  
IP Address . . . . . : 10.0.0.10  
Subnet Mask . . . . . : 255.0.0.0  
Default Gateway . . . . . :  
  
C:\>ping 10.0.0.10  
  
Pinging 10.0.0.10 with 32 bytes of data:  
  
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128  
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128  
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128  
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128  
  
Ping statistics for 10.0.0.10:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 0ms, Maximum = 0ms, Average = 0ms  
  
C:\>ping 10.0.0.20  
  
Pinging 10.0.0.20 with 32 bytes of data:  
  
Reply from 10.0.0.20: bytes=32 time=4ms TTL=128  
Reply from 10.0.0.20: bytes=32 time=2ms TTL=128  
Reply from 10.0.0.20: bytes=32 time=2ms TTL=128  
Reply from 10.0.0.20: bytes=32 time=2ms TTL=128  
  
Ping statistics for 10.0.0.20:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 2ms, Maximum = 4ms, Average = 2ms  
  
C:\>ping 20.0.0.10  
  
Pinging 20.0.0.10 with 32 bytes of data:  
  
Destination host unreachable.  
Destination host unreachable.  
Destination host unreachable.  
Destination host unreachable.  
  
Ping statistics for 20.0.0.10:  
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),  
  
C:\>ping 20.0.0.20  
  
Pinging 20.0.0.20 with 32 bytes of data:  
  
Destination host unreachable.  
Destination host unreachable.  
Destination host unreachable.  
Destination host unreachable.  
  
Ping statistics for 20.0.0.20:  
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),  
  
C:\>

Go

StartC:\WINDOWS\system32\cmd.exeNetwork Connections

31°C Mostly sunnySearch

10:26 AM 16-09-2026

PNETLab | Topology

VLAN.cdr

Terminal

labs.surennetwork.com/store/public/admin/labs/terminal?terminals=eyJ0YWVzljpbeyJuYW11IjoieV2luLTEiLCJ1cmwiOiIvaHRtbDUvly9jbGllbnQvTXpFek16UXINREk0QUdNQWJYbHJpV3c9P3Rva2VuPTBGOTewQTIDOTQ0RDQ0OTYz...

Incognito (2)

Finish update

Win-1Win-2Win-3Win-4SW1SW2

C:\WINDOWS\system32\cmd.exe

Ethernet adapter Local Area Connection:  
  
Connection-specific DNS Suffix . :  
IP Address . . . . . : 20.0.0.10  
Subnet Mask . . . . . : 255.0.0.0  
Default Gateway . . . . . :  
  
C:\>ping 20.0.0.10  
  
Pinging 20.0.0.10 with 32 bytes of data:  
  
Reply from 20.0.0.10: bytes=32 time<1ms TTL=128  
Reply from 20.0.0.10: bytes=32 time<1ms TTL=128  
Reply from 20.0.0.10: bytes=32 time<1ms TTL=128  
Reply from 20.0.0.10: bytes=32 time<1ms TTL=128  
  
Ping statistics for 20.0.0.10:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 0ms, Maximum = 0ms, Average = 0ms  
  
C:\>ping 20.0.0.20  
  
Pinging 20.0.0.20 with 32 bytes of data:  
  
Reply from 20.0.0.20: bytes=32 time=4ms TTL=128  
Reply from 20.0.0.20: bytes=32 time=2ms TTL=128  
Reply from 20.0.0.20: bytes=32 time=1ms TTL=128  
Reply from 20.0.0.20: bytes=32 time=2ms TTL=128  
  
Ping statistics for 20.0.0.20:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 1ms, Maximum = 4ms, Average = 2ms  
  
C:\>ping 10.0.0.10  
  
Pinging 10.0.0.10 with 32 bytes of data:  
  
Destination host unreachable.  
Destination host unreachable.  
Destination host unreachable.  
Destination host unreachable.  
  
Ping statistics for 10.0.0.10:  
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),  
  
C:\>ping 10.0.0.20  
  
Pinging 10.0.0.20 with 32 bytes of data:  
  
Destination host unreachable.  
Destination host unreachable.  
Destination host unreachable.  
Destination host unreachable.  
  
Ping statistics for 10.0.0.20:  
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),  
  
C:\>

Start

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Search

10:27 AM 16-09-2026

PNETLab | Topology

VLAN.cdr

Terminal

labs.surennetwork.com/store/public/admin/labs/terminal?terminals=eyJ0YWVzljpbeyJuYW11IjoieV2luLTEiLCJ1cmwiOiIvaHRtbDUvly9jbGllbnQvTXpFek16UXINREk0QUdNQWJYbHpwV3c9P3Rva2VuPTBGOTExQTIDOTQ0RDQ0OTYz...

Incognito (2)

Finish update

Win-1Win-2Win-3Win-4SW1SW2

C:\WINDOWS\system32\cmd.exe

Ethernet adapter Local Area Connection:  
Connection-specific DNS Suffix . :  
IP Address . . . . . : 10.0.0.20  
Subnet Mask . . . . . : 255.0.0.0  
Default Gateway . . . . . :  
  
C:\>ping 10.0.0.20  
  
Pinging 10.0.0.20 with 32 bytes of data:  
  
Reply from 10.0.0.20: bytes=32 time<1ms TTL=128  
Reply from 10.0.0.20: bytes=32 time<1ms TTL=128  
Reply from 10.0.0.20: bytes=32 time<1ms TTL=128  
Reply from 10.0.0.20: bytes=32 time<1ms TTL=128  
  
Ping statistics for 10.0.0.20:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 0ms, Maximum = 0ms, Average = 0ms  
  
C:\>ping 10.0.0.10  
  
Pinging 10.0.0.10 with 32 bytes of data:  
  
Reply from 10.0.0.10: bytes=32 time=3ms TTL=128  
Reply from 10.0.0.10: bytes=32 time=3ms TTL=128  
Reply from 10.0.0.10: bytes=32 time=2ms TTL=128  
Reply from 10.0.0.10: bytes=32 time=1ms TTL=128  
  
Ping statistics for 10.0.0.10:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 1ms, Maximum = 3ms, Average = 2ms  
  
C:\>ping 20.0.0.10  
  
Pinging 20.0.0.10 with 32 bytes of data:  
  
Destination host unreachable.  
Destination host unreachable.  
Destination host unreachable.  
Destination host unreachable.  
  
Ping statistics for 20.0.0.10:  
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),  
  
C:\>ping 20.0.0.20  
  
Pinging 20.0.0.20 with 32 bytes of data:  
  
Destination host unreachable.  
Destination host unreachable.  
Destination host unreachable.  
Destination host unreachable.  
  
Ping statistics for 20.0.0.20:  
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),  
  
C:\>

Start

C:\WINDOWS\system...

5:59 AM

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10:29 AM 16-09-2026

PNETLab | Topology

VLAN.cdr

Terminal

labs.surennetwork.com/store/public/admin/labs/terminal?terminals=eyJ0YWVzljpbeyJuYW11lloiV2luLTEiLCJ1cmwiOiIvaHRtbDUvly9jbGllbnQvTXpFek16UXINREk0QUdNQWJYbHJpV3c9P3Rva2VuPTBGOTewQTIDOTQ0RDQ0OTYz...

Incognito (2)

Finish update

Win-1 x Win-2 x Win-3 x Win-4 x SW1 x SW2 x

C:\WINDOWS\system32\cmd.exe

Ethernet adapter Local Area Connection:  
Connection-specific DNS Suffix . :  
IP Address . . . . . : 20.0.0.20  
Subnet Mask . . . . . : 255.0.0.0  
Default Gateway . . . . . :  
C:\>ping 20.0.0.20  
Pinging 20.0.0.20 with 32 bytes of data:  
Reply from 20.0.0.20: bytes=32 time<1ms TTL=128  
Reply from 20.0.0.20: bytes=32 time<1ms TTL=128  
Reply from 20.0.0.20: bytes=32 time<1ms TTL=128  
Reply from 20.0.0.20: bytes=32 time<1ms TTL=128  
Ping statistics for 20.0.0.20:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 0ms, Maximum = 0ms, Average = 0ms  
C:\>ping 20.0.0.10  
Pinging 20.0.0.10 with 32 bytes of data:  
Reply from 20.0.0.10: bytes=32 time=3ms TTL=128  
Reply from 20.0.0.10: bytes=32 time=2ms TTL=128  
Reply from 20.0.0.10: bytes=32 time=1ms TTL=128  
Reply from 20.0.0.10: bytes=32 time=1ms TTL=128  
Ping statistics for 20.0.0.10:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 1ms, Maximum = 3ms, Average = 1ms  
C:\>ping 10.0.0.10  
Pinging 10.0.0.10 with 32 bytes of data:  
Destination host unreachable.  
Destination host unreachable.  
Destination host unreachable.  
Destination host unreachable.  
Ping statistics for 10.0.0.10:  
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),  
C:\>ping 10.0.0.20  
Pinging 10.0.0.20 with 32 bytes of data:  
Destination host unreachable.  
Destination host unreachable.  
Destination host unreachable.  
Destination host unreachable.  
Ping statistics for 10.0.0.20:  
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),  
C:\>

Start

C:\WINDOWS\system...

5:59 AM

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Search

10:30 AM 16-09-2026

## Keywords

**VLAN Trunking:** The concept of carrying traffic from multiple VLANs over a single physical link between SW1 and SW2 (e0/2).

**VLAN Segmentation:** Isolating devices into distinct broadcast domains (VLAN 10 and VLAN 20).

**Access Ports:** The switch ports directly connecting to the Windows PCs (e0/0, e0/1), configured to handle only one specific VLAN's traffic.

**802.1Q Encapsulation:** The standard protocol used to tag Ethernet frames traversing the trunk link so the receiving switch knows which VLAN they belong to.

**Subnetting:** The IP addressing scheme separating the networks (10.0.0.0/x and 20.0.0.0/x).

**1. If the link between SW1 and SW2 is configured as a trunk port, which PCs will be able to communicate with each other?**

**Answer:** Win-1 can communicate with Win-3, and Win-2 can communicate with Win-4.

**Reasoning:** Trunk links allow traffic from both VLAN 10 and VLAN 20 to pass between the switches. Devices within the same VLAN and same IP subnet can communicate natively.

**2. Can Win-1 (10.0.0.10) ping Win-2 (20.0.0.10) through this current setup?**

**Answer:** No.

**Reasoning:** Win-1 is on **VLAN 10** and Win-2 is on **VLAN 20**. Switches operate at Layer 2 and cannot route traffic between different VLANs/subnets on their own. Inter-VLAN routing (like a Router-on-a-Stick or a Layer 3 Switch) is required for them to talk.

**3. What happens to a frame sent from Win-1 as it travels across the link between SW1 and SW2?**

**Answer:** SW1 receives an untagged frame from Win-1 on its access port (e0/0). As the frame leaves SW1 via the trunk port (e0/2), **SW1 adds an 802.1Q VLAN tag (ID: 10)** to it. SW2 reads this tag, strips it off, and forwards the clean frame out of its own VLAN 10 access port (e0/0) to Win-3.

## 5. What is VLAN (Virtual Local Area Network)?

A **VLAN** is a custom network created by logically grouping together a subset of devices on a switch, regardless of their physical location.

**How it works:** In your diagram, you have two VLANs: **VLAN 10** (purple boxes) and **VLAN 20** (green boxes).

**Core Benefit:** It improves security and reduces unnecessary network traffic (broadcast storms) by keeping different departments or system types isolated.

## 6. What is a Trunk?

A **Trunk** (or trunk link) is a single physical connection between two network devices (like the wire connecting port e0/2 on SW1 to SW2) that is configured to carry traffic for **multiple VLANs** simultaneously.

**How it works:** Without a trunk, you would need to run one physical cable between the switches for VLAN 10, and a second physical cable for VLAN 20. A trunk eliminates this by adding a hidden "tag" (using the **802.1Q standard**) to the data frames as they cross the wire, telling the receiving switch exactly which VLAN the data belongs to.

**Core Benefit:** It saves hardware costs, reduces cable clutter, and makes scaling the network much easier.